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Non-invasive classification of IDH mutation status of gliomas from multi-modal MRI using a 3D convolutional neural network

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Abstract

Glioma is the most common form of brain tumor with a high degree of heterogeneity in imaging characteristics, treatment-response, and survival rate. An important factor causing this heterogeneity is the mutation of isocitrate dehydrogenase (IDH) enzyme. The current clinical gold-standard for identifying IDH mutation status involves invasive procedures that involve risk, may fail to capture intra-tumoral spatial heterogeneity or can be inaccessible in low-resource settings. In this study, we propose a deep learning-based method to non-invasively and pre-operatively determine IDH status of high- and low-grade gliomas by leveraging their phenotypical characteristics from volumetric MRI scans. For this purpose, we propose a 3D Mask R-CNN-based approach to simultaneously detect and segment glioma as well as classify its IDH status - thus obviating the requirement of any separate tumor segmentation step. The network can operate on routinely acquired MRI sequences and is agnostic to glioma grade. It was trained on patient-cases from publicly available datasets (n = 223) and tested on two hold-out datasets acquired from The Cancer Genome Atlas (TCGA; n = 62) and Washington University School of Medicine (WUSM; n = 261). The model achieved areas under the receiver operating characteristic of 0.83 and 0.87, and areas under the precision-recall curves of 0.78 and 0.79, on the TCGA and WUSM sets, respectively. The model can be used to perform a pre-operative ‘virtual biopsy’ of gliomas, thus facilitating treatment planning, potentially leading to better overall survival.

Keywords

glioma; Isocitrate Dehydrogenase; convolutional neural network; Mask R-CNN; deep learning; neuro-oncology; tumor classification; tumor segmentation; tumor detection

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1. INTRODUCTION

Glioma is the most common form of brain tumor with a high degree of heterogeneity in its imaging characteristics, response to therapy, and survival rate. An important factor causing this heterogeneity is the mutation of isocitrate dehydrogenase (IDH) enzyme. Therefore, an accurate pre-operative determination of IDH status can significantly assist in treatment planning and improve treatment outcome.

The current clinical gold standard for identifying IDH mutation status involves invasive brain biopsy procedures. These procedures have associated risks,[1] may fail to capture intra-tumoral spatial heterogeneity due to localized samples, or may be inaccessible in low-resource settings. So, MRI can be used as a potential alternative due to its non-invasive nature, easy accessibility, high soft-tissue contrast, and its ability to capture a holistic view of the tumor.

In the last few years, various machine- (ML) and deep-learning (DL) based approaches[2] have attempted to classify IDH mutation status by leveraging the variation in phenotypical characteristics of gliomas that are manifested in MRI scans due to changes in IDH mutations.[3] However, despite these efforts, most of these methods are limited by the requirement of tumor segmentation masks (either manually drawn[4–6] or separately generated using a second DL model[3, 7, 8]), lack of external validation data[3, 9] necessary for assessing the generalizability of models, or applicability on only specific grades of glioma (e.g., only low grade[4, 10]). Additionally, studies using a combination of ML and radiomic features to classify IDH status suffer from requirements of manual feature selection and feature engineering associated with ML as well as repeatability and reproducibility issues associated with radiomic features.[11]

To address these limitations, we have proposed a 3D convolutional neural network (CNN) approach for classifying IDH mutation status of gliomas into IDH-wildtype (IDH-wt) and IDH-mutant (IDH-mut) classes. We have included all grades of glioma (WHO 2021 grades 2-4) to facilitate an easier translation of this model to a pre-operative clinical setting. Our approach works on routinely acquired structural MR sequences without the requirement of any prior tumor segmentation, thus significantly simplifying the workflow compared to most previous studies. We have validated our model on two independent hold-out sets acquired from public and clinical sources to demonstrate its generalizability.

2. METHODS

2.1 3D Mask R-CNN architecture

For classification, we adopted a 3D Mask R-CNN architecture[12] (Fig. 1) with a Residual Network (ResNet-101)-Feature Pyramid Network (FPN) backbone.[13, 14] The choice of Mask R-CNN is motivated by its capability of jointly classifying, detecting, and segmenting the objects of interest. Although classification is the main goal of the study, the two auxiliary tasks of detection and segmentation serve two purposes. First, they obviate the requirement of any prior glioma segmentation step. Second, the network first accurately localizes the glioma and then classifies its molecular status, without being influenced by confounding

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factors like image artifacts. The 3D network is capable of classifying entire MR volumes in a single forward pass. During testing, the network detects the tumor using a bounding box, segments the whole tumor, and classifies its IDH status (Fig. 2).

2.2 Hyperparameters and training strategy

The network was fed with 128 × 128 × 128 volumes from each subject's post-contrast T1-weighted (T1c), T2-weighted (T2), and T2-weighted Fluid-attenuated inversion recovery (FLAIR) scans. The best hyperparameters were determined using five-fold cross-validation. The data was split at the subject level to prevent any data leakage between different subsets of data. The model was trained for 200 epochs with a batch-size of 4, a learning rate of 1×10^{-4}, and Adam optimizer. To prevent overfitting, training-time data-augmentations of mirroring and random rotations were performed. To address class-imbalance, we used a class-weighted softmax cross-entropy loss in the classification head of the model. The code was implemented in Python 3.6.2, using the PyTorch 0.4.1 library. All experiments were performed on an NVIDIA TESLA V100S-32GB GPU. The code and trained models are available at https://github.com/satrajitgithub/glioma_molecular_3D.

2.3 Datasets and pre-processing

The model was trained and cross-validated on 223 patient-cases from the Brain Tumor Segmentation challenge[15–17] (BraTS; n = 189), and Ivy Glioblastoma Atlas Project[18] (Ivy GAP; n = 34) datasets (Tab. 1). Two external datasets were built to assess model generalizability viz. (i) TCGA (n = 62) comprising data from The Cancer Genome Atlas (TCGA-GBM[19] and TCGA-LGG[20]), and (ii) WUSM (n = 261) comprising retrospective data acquired from the Washington University School of Medicine.

Scans from BraTS and Ivy GAP datasets were already registered to the SRI24 anatomical atlas,[21] resampled to $1mm^3$ isotropic resolution and skull-stripped. To bring all data to the same anatomical space, raw data from TCGA and WUSM were pre-processed using the same protocol as BraTS using the Integrative Imaging Informatics for Cancer Research: Workflow Automation for Neuro-oncology (I3CR-WANO) framework.[22,23] Pre-processing included conversion of all DICOM scans into NIfTI format, reorientation of all scans to the left-posterior-superior (LPS) orientation, rigid co-registration of T2 and FLAIR scans to the T1c scan on a per-subject basis, rigid registration of T1c to the SRI24 atlas to obtain the affine transformation from patient space to atlas space, and finally applying this transformation to all co-registered images. After registration, all scans were N4 bias-corrected and skull-stripped using the Robust Brain Extraction (ROBEX)[24] tool. Once scans across all included datasets were in the same anatomical space, image intensities within the brain were normalized to zero mean and unit variance after excluding intensities below the 5th and above the 95th percentile.

2.4 Comparisons to baselines

The classification performance of the proposed model was compared to two baseline methods: (i) a cascaded approach proposed by Choi et al.[8] (hereon referred to as "Choi-CNN"), where the authors first use a network to segment the glioma and then use a second network to classify its IDH status, and (ii) a multi-task U-net model by Voort et al.[25]

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(hereon referred to as “Voort-CNN”), where the authors perform segmentaion of glioma and classification of molecular subtype within the same network.

All comparisons were performed on the WUSM set. The Choi-CNN model required the same sequences as the proposed model. Results for Choi-CNN were obtained using the pre-trained model provided by the authors (https://github.com/yoonchoi-neuro/automated_hybrid_IDH). Results for Voort-CNN were obtained using the pre-trained model provided by the authors as the svdvoort/prognosais_glioma:1.0.2 docker. In addition to T1c, T2, and FLAIR sequences, the Voort-CNN model required pre-contrast T1-weighted (T1w) MRI scans, which were available for all 261 cases from the WUSM dataset.

2.5 Statistical analysis

We used Chi-square and Mann-Whitney tests to evaluate differences in patient demographics and clinical characteristics between data splits. The classification performance was quantified using accuracy, precision, recall, F1 score, area under receiver operating characteristics (AUROC), and area under precision-recall curves (AUPRC). For AUROC and AUPRC, 95% confidence intervals (CI) were calculated using a 1000-sample bootstrapping method, i.e., by resampling the prediction scores 1000 times with replacement, calculating the scores for each of those 1000 instances and taking the 5th and 95th percentile values from the sample of 1000 values. Confusion matrices were calculated to show the error distribution across different classes. Statistical comparisons between proposed and baseline methods were performed using the Mcnemar test[26] for precision, the generalized score statistic[27] for recall, and the DeLong test[28] for AUROCs. All statistical analyses and visualizations of the results were performed using scikit-learn, numpy, pandas, and seaborn libraries in Python 3.6.2, and the DTComPair and pROC packages in R, version 4.1.0 (https://www.rproject.org/). For all statistical analyses, the threshold for statistical significance was set to $P < .05$.

3. RESULTS

The model predicted IDH status (Tab. 2, Fig. 3) with high accuracy across both TCGA (80.6%) and WUSM (84.3%) sets. Precision, recall, AUROC, and AUPRC values were also similar for both external sets. Overall, the model showed good generalization, with the performance being slightly higher in WUSM compared to TCGA. Compared to the Choi-CNN baseline (“WUSM-Choi” shown in Tab. 2, Fig. 3), the proposed model performed significantly better in terms of precision (0.21 increase, $P < 0.001$) and AUROC (0.125 increase, $P = 0.006$). Compared to the Voort-CNN baseline (“WUSM-Voort” shown in Tab. 2, Fig. 3), the proposed model performed significantly better in terms of precision (0.23 increase; $P < 0.001$), recall (0.188 increase; $P = 0.012$), and AUROC (0.117 increase; $P = 0.002$).

4. DISCUSSION

In this work, we proposed a 3D Mask R-CNN network for the classification of IDH mutation status. Our main contributions are the following. First, our model jointly detects and segments the glioma before classifying its molecular status. This is contrary to all

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previous ML-based studies that required prior tumor segmentation and most DL-based studies that required multiple task-specific models (e.g., a cascade of tumor segmentation model with molecular classification model). This advantage obviates the requirement for tedious manual annotation or additional tumor segmentation models, thus resulting in a streamlined workflow. Additionally, the usage of a multi-task model helps with learning shared information between multiple related tasks of detection and classification. Second, as the proposed approach is 3D, it does not require any prior manual selection of the best slice (typically, with the largest tumor area) to be fed as input into 2D ML or DL models. Third, our model is agnostic to glioma grade, which makes its translation to a pre-operative setting much simpler than grade-specific models. The proposed method showed good generalization on both clinical and public external datasets and performed significantly better than the two baseline models on the clinical set.

The current study showed promising results in classifying IDH status from T1c, T2, and FLAIR sequences. This work can be expanded in two main directions in the future. First, future works will explore the feasibility of a comprehensive classification of gliomas as per the WHO 2021 classification system.[29] To achieve this, we require the knowledge of both IDH mutation status and co-deletion status of chromosome arms 1p/19q to classify gliomas as “Astrocytoma, IDH-mutant” and “Oligodendroglioma, IDH-mutant and 1p/19q-codeleted”. In addition, we require the knowledge of TERT promoter mutation, EGFR gene amplification, and combined gain of entire chromosome 7 and loss of entire chromosome 10 (+7/−10) to classify IDH-wt, grade 2/3 gliomas as “Glioblastoma, IDH-wildtype” and “NEC (not elsewhere classified)”. Accordingly, we plan to explore whether a DL model can simultaneously predict the aforementioned molecular factors. Second, although the usage of routinely acquired structural sequences simplifies the clinical translation of the proposed method, other advanced sequences like perfusion imaging or MR-spectroscopy can be explored.

In conclusion, we developed a CNN model to classify IDH mutation status from pre-operative MR volumes. Compared to invasive assays used as current clinical gold-standard for molecular assessment, the proposed workflow offers a faster, more cost-effective, and non-invasive solution for pre-operative determination of IDH status that can support clinical decision-making and lead to a better treatment outcome. In this emerging era of precision diagnostics, this workflow can drive personalized treatment planning by streamlining molecular characterization of gliomas. The proposed method can be combined with previous methods developed for classifying various intracranial tumors[30] into a comprehensive brain tumor classification system.

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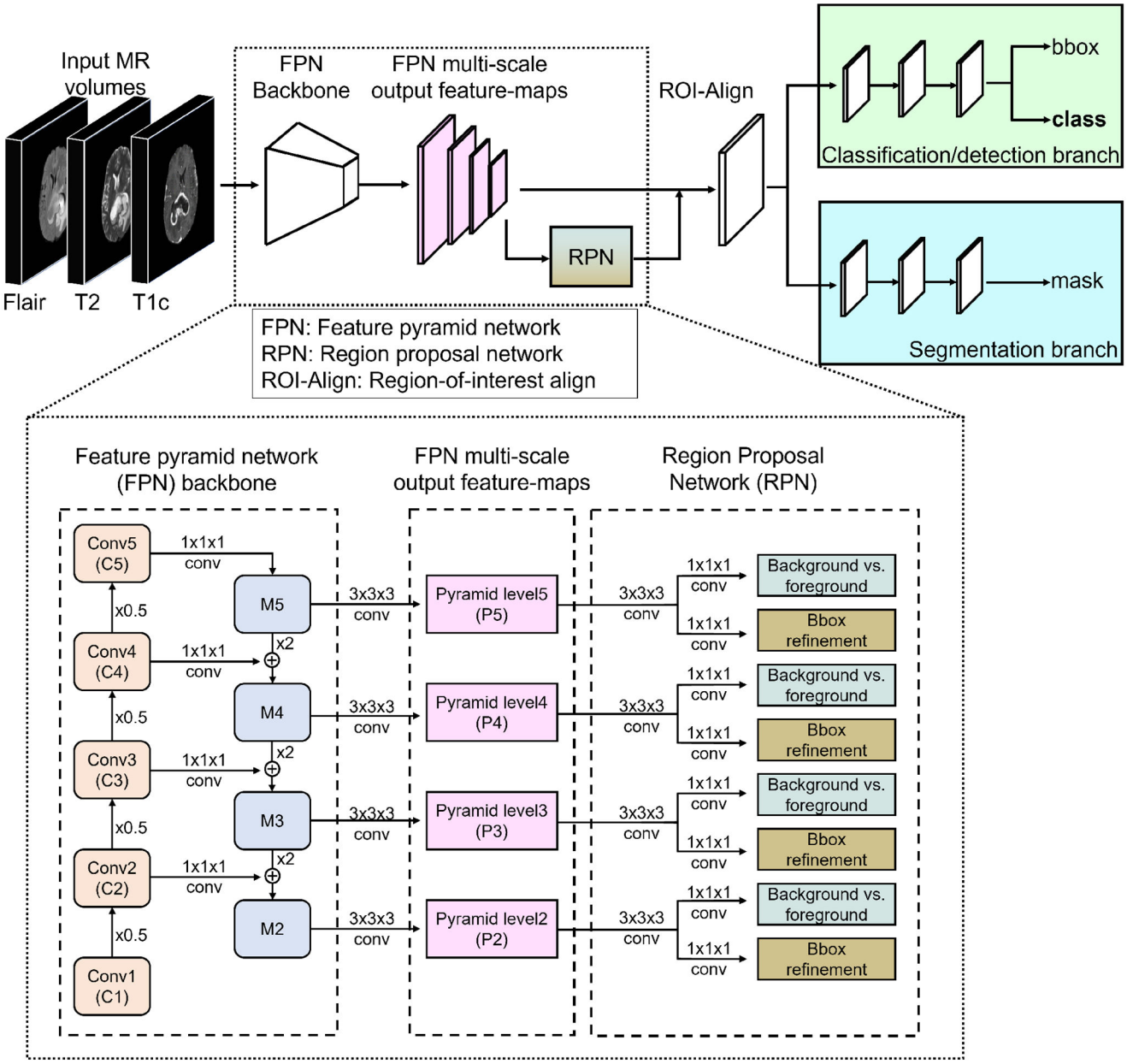


Figure 1.

Schematic of the proposed 3D Mask R-CNN for predicting IDH status including the detailed views of the Feature Pyramid Network (FPN) backbone and the Region Proposal Network (RPN). The model takes as input three MR volumes of T1c, T2, and FLAIR sequences and produces a classification result of IDH-wt vs. IDH-mut at the “class” head of the network within the classification/detection branch.

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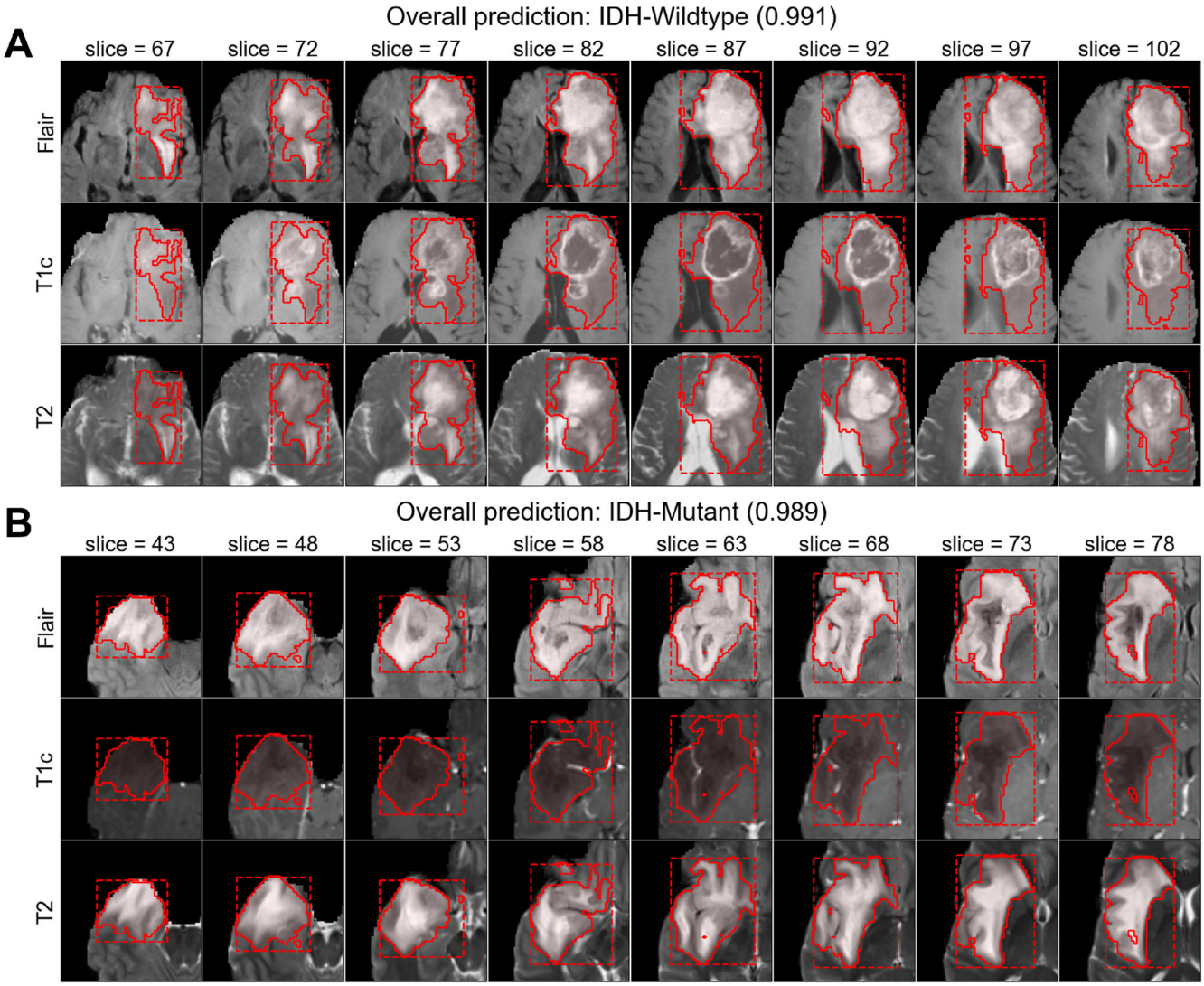


Figure 2.

Example model predictions shown slice-by-slice for ease of visualization for (A) an IDH-wt glioma (shown between slices 67-102 in steps of 5), and (B) an IDH-mut glioma (shown between slices 43-78 in steps of 5). In both cases, the figures show the classification result (as “Overall prediction”), the detected tumor (with a bounding box drawn with red dashed line), and the segmented tumor (shown using red mask overlay).

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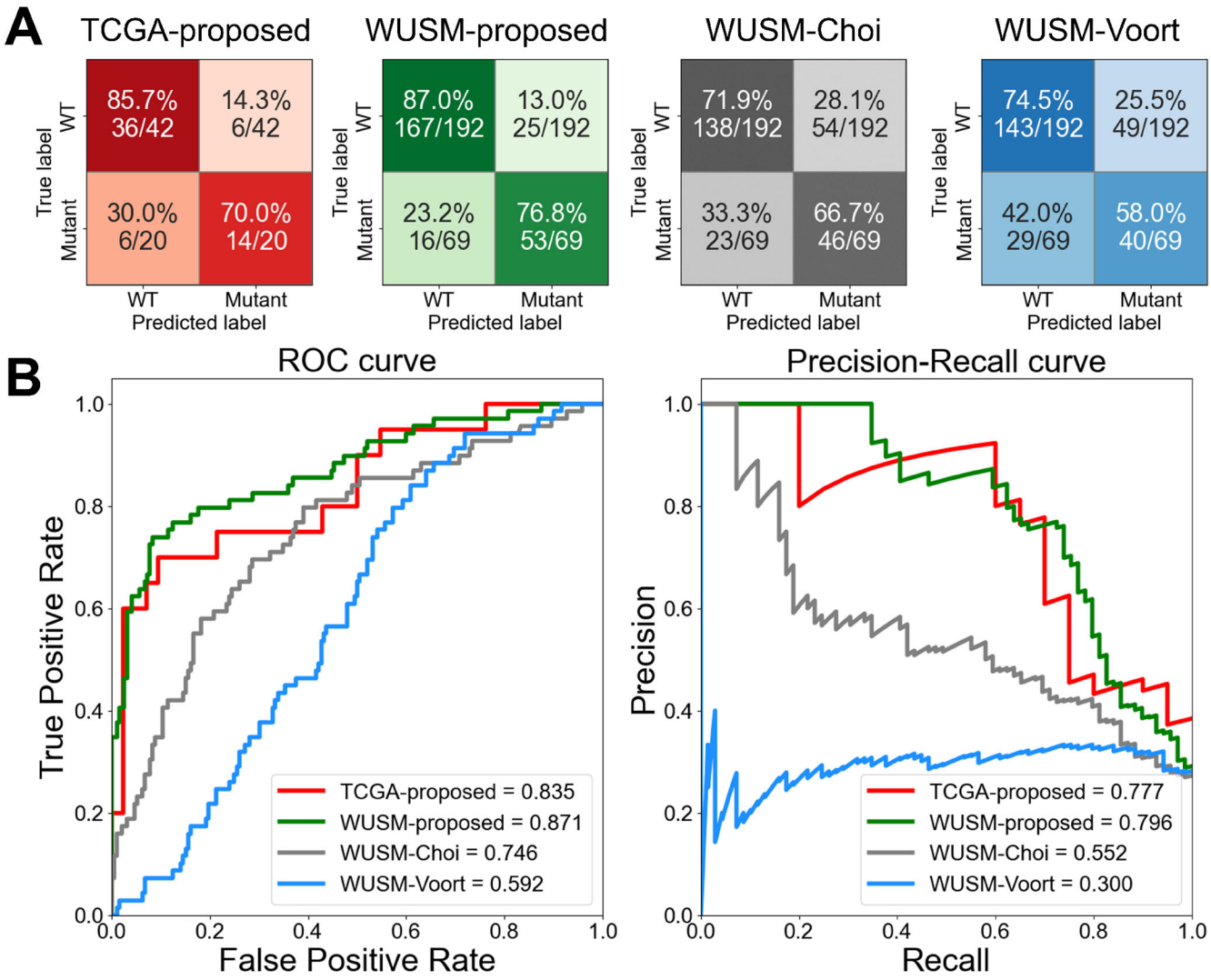


Figure 3.

(A) Confusion matrix for results of the proposed method on TCGA and WUSM test sets (TCGA-proposed, WUSM-proposed), and two baseline methods on the WUSM test set (WUSM-Choi, WUSM-Voort) (B) Receiver Operating Characteristics and Precision-Recall curves for TCGA-proposed, WUSM-proposed, WUSM-Choi, and WUSM-Voort.

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Table 1.

Patient characteristics for the training, TCGA, and WUSM splits of data.

	training (n = 223)	**TCGA (n = 62)**	**P-value (TCGA)**	**WUSM (n = 261)**	**P-value (WUSM)**
Age	54 (41 - 62)	58 (49 - 66)	0.041	58 (44 - 67)	0.002
Sex			0.4838		0.0079
Female	107 (47.98%)	26 (41.94%)		93 (35.63%)	
Male	116 (52.02%)	36 (58.06%)		168 (64.37%)	
IDH status			0.2829		0.0011
Mutant	91 (40.81%)	20 (32.26%)		69 (26.44%)	
Wildtype	132 (59.19%)	42 (67.74%)		192 (73.56%)	
WHO Grade			<0.001		<0.001
2	47 (21.08%)	11 (17.74%)		56 (21.46%)	
3	59 (26.46%)	5 (8.06%)		37 (14.18%)	
4	104 (46.64%)	46 (74.19%)		166 (63.6%)	
Unknown	13 (5.83%)			2 (0.77%)	

Parenthetical values represent the percentage of each subgroup within a specific dataset (i.e., training, TCGA, WUSM).

For the ‘Age’ row, interquartile range (IQR) is given in parentheses.

P-values demonstrate differences between training data vs. the dataset mentioned in parenthesis in the header row.

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Table 2.

Performance of the proposed model on TCGA and WUSM test sets (TCGA-proposed, WUSM-proposed), and the two baseline models on the WUSM test set (WUSM-Choi, WUSM-Voort)

	Accuracy	Precision	Recall	F1-score	AUROC (95% CI)	AUPRC (95% CI)
TCGA-proposed	0.806	0.7	0.7	0.7	0.834 (0.71, 0.94)	0.776 (0.60, 0.93)
WUSM-proposed	0.843	0.679	0.768	0.721	0.871 (0.82, 0.92)	0.796 (0.714, 0.873)
WUSM-Choi	0.705	0.46	0.667	0.544	0.746 (0.675, 0.812)	0.552 (0.434, 0.668)
WUSM-Voort	0.701	0.449	0.579	0.506	0.591 (0.522, 0.668)	0.299 (0.237, 0.402)